

Project Report: Limit-Track Token Manager & Extension Sync

Project Name: Limit-Track
Version: 1.0.0 (Odoo 19.0 Compatible)
Developer: Safwan Ahmad Saffi
Status: Completed & Production-Ready

1. Executive Summary

The **Limit-Track** system is a unified solution designed to bridge the gap between local browser-based LLM (Large Language Model) usage and a centralized management dashboard. It consists of a custom **Odoo 19.0 Module** and a **Manifest V3 Chrome Extension**. The system provides real-time token tracking, automated balance synchronization, and detailed usage analytics.

2. System Architecture

The project follows a **Client-Server architecture** where the Odoo instance acts as the secure central "Brain" and the Chrome Extension acts as the automated "Sensor."

Component	Technology Stack	Primary Function
Backend	Odoo 19.0 (Python, XML, PostgreSQL)	Manages accounts, token balances, and secure API keys.
Frontend	Chrome Extension (JS, HTML, CSS)	Monitors LLM platforms (ChatGPT, Claude, Gemini) and syncs usage.
API Layer	JSON-RPC 2.0 / REST	Facilitates secure, real-time communication between components.
Hosting	SkySize (Managed Odoo Hosting)	Provides the cloud infrastructure for the backend.

3. Odoo Module Implementation (skysize_token_manager)

3.1 Data Models

- **skysize.token.account** : Stores user-specific data, including the current token balance, max limits, and the secure **UUID Access Token**.
- **skysize.token.log** : A transactional log that records every token deduction, including a timestamp, amount used, and a reference to the specific LLM platform.

3.2 Key Features

- **Odoo 19 List View**: Fully optimized for the new Odoo 19 **list** architecture for high performance.
 - **Secure API Authentication**: Each user account generates a unique UUID4 token that must be passed in every API request.
 - **Automated Deduction Logic**: A custom Python method handles balance validation and prevents "overdrafts" (using tokens beyond the available balance).
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4. Extension Integration & Auto-Sync

The extension is designed to be "zero-touch" for the user once configured.

- **Automatic Token Estimation**: Uses a calibrated heuristic to estimate tokens across different platforms (ChatGPT, Claude, DeepSeek, Gemini).
 - **Settings Dashboard**: A dedicated popup UI allows the user to securely input their Odoo instance URL and Access Token.
 - **Background Sync**: A service worker handles the API communication in the background, ensuring the browser UI never freezes during synchronization.
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5. Deployment Workflow

The project was deployed using a modern **GitOps workflow**:

1. **Repository Integration**: Linked the GitHub repository **safwanahmadsaffi/limit-track** to the SkySize instance.
 2. **Continuous Deployment**: Every code update (including the Odoo 19 **list** view fixes) was automatically built and deployed to **limit-track.skysize.io**.
 3. **Module Activation**: The module was activated via the Odoo Developer Tools, successfully initializing the database schema and menu structures.
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6. Challenges Faced & Solution Strategy

During the development and deployment of the Limit-Track project, several technical hurdles were encountered and resolved:

6.1 Odoo 19.0 Compatibility Issues

- **Problem:** The initial module activation failed due to strict view validation in Odoo 19. Specifically, the legacy `<tree>` tag was deprecated and resulted in an `RPC_ERROR` during installation.
- **Strategy:** Performed a deep-dive audit of the Odoo 19 framework changes. The solution involved migrating all XML view architectures from the deprecated `tree` tag to the new `list` tag, ensuring compliance with the latest Odoo standards.

6.2 Real-Time Token Synchronization

- **Problem:** Ensuring the browser extension could communicate securely with a public Odoo instance without exposing user credentials.
- **Strategy:** Implemented a UUID-based token authentication layer. The Odoo backend generates unique "Magic Keys" for each user, allowing the extension to securely deduct tokens via a specialized JSON-RPC controller (`/api/skysize/tokens/consume`) without requiring a full Odoo login.

6.3 Deployment & Build Automation

- **Problem:** Coordinating updates between the GitHub repository and the SkySize managed instance during the rapid bug-fixing phase.
- **Strategy:** Adopted a forced-push GitOps strategy to trigger immediate rebuilds on SkySize. This ensured that the Odoo "Apps List" could be updated in real-time to reflect the latest compatibility fixes.

7. Conclusion

The **Limit-Track** project successfully delivers a professional-grade token management system. By integrating Odoo's powerful ERP capabilities with a lightweight browser extension, the system provides a scalable and secure way to monitor and manage LLM resources.

Final Deliverables Included:

- Complete Odoo 19.0 Source Code (GitHub)
- Pre-built Chrome Extension (`dist/` folder)

- Unified Project Documentation & Integration Guide
 - Production ZIP Archive
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8. Project Access Links

For evaluation and live demonstration, the project can be accessed via the following links:

- **GitHub Repository:** <https://github.com/safwanahmadsaffi/limit-track>
 - **Live Odoo Instance:** <https://limit-track.skysize.io/web>
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